

Menu-Driven Program for Digit Operations

Aim: To write a menu-driven C program using switch-case to perform operations on the digits of a positive integer using while, for, and do-while loops.

Algorithm

1. Start.
2. Input a positive integer n.
3. Display the menu: (1) Sum of squares of digits, (2) Sum of cubes of digits, (3) Difference between sum of cubes and sum of squares.
4. Input the user's choice.
5. Use switch-case to perform the selected operation.
6. For choice 1, use a while loop to extract each digit, square it, and add it to square_sum.
7. For choice 2, use a for loop to extract each digit, cube it, and add it to cube_sum.
8. For choice 3, use a do-while loop to calculate both sums and find Difference = cube_sum - square_sum.
9. Display the result.
10. If the choice is invalid, display 'Invalid choice'.
11. Stop.

Test Case Table

Test	Number	Choice	Operation	Expected Output
1	123	1	Sum of squares	14
2	123	2	Sum of cubes	36
3	123	3	Difference	22
4	25	1	Sum of squares	29
5	25	2	Sum of cubes	133
6	25	3	Difference	104
7	123	4	Invalid choice	Invalid choice

Verification for Number 123

Sum of squares = $1^2 + 2^2 + 3^2 = 1 + 4 + 9 = 14$

Sum of cubes = $1^3 + 2^3 + 3^3 = 1 + 8 + 27 = 36$

Difference = $36 - 14 = 22$

C Program

```
#include <stdio.h>

int main()
{
    int n, temp, digit, choice;
    int square_sum = 0, cube_sum = 0, difference;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    if(n <= 0)
    {
        printf("Please enter a positive integer.");
        return 0;
    }

    printf("\n1. Sum of squares of digits");
    printf("\n2. Sum of cubes of digits");
    printf("\n3. Difference between sum of cubes and squares");
    printf("\nEnter your choice: ");
    scanf("%d", &choice);

    temp = n;

    if(choice == 1)
    {
        while(temp > 0)
        {
            digit = temp % 10;
            square_sum = square_sum + digit * digit;
            temp = temp / 10;
        }

        printf("Sum of squares = %d", square_sum);
    }

    else if(choice == 2)
    {
        for(; temp > 0; temp = temp / 10)
        {
            digit = temp % 10;
            cube_sum = cube_sum + digit * digit * digit;
        }

        printf("Sum of cubes = %d", cube_sum);
    }

    else if(choice == 3)
    {
        do
        {
            digit = temp % 10;

            square_sum = square_sum + digit * digit;
            cube_sum = cube_sum + digit * digit * digit;

            temp = temp / 10;
        } while(temp > 0);

        difference = cube_sum - square_sum;
        printf("Difference between sum of cubes and squares = %d", difference);
    }
}
```

```
    } while(temp > 0);

    difference = cube_sum - square_sum;

    printf("Difference = %d", difference);
}

else
{
    printf("Invalid choice");
}

return 0;
}
```

Compilation and Execution

```
gcc program.c
./a.out
```

Sample Test Runs

Test 1:

Enter a positive integer: 123 Enter

your choice: 1

Sum of squares = 14

Test 2:

Enter a positive integer: 123 Enter

your choice: 2

Sum of cubes = 36

Test 3:

Enter a positive integer: 123 Enter

your choice: 3 Difference = 22

Result

The menu-driven C program was prepared, compiled, executed, and tested for the required operations. The obtained results match the prepared test cases.